



TripPilot

Your AI Co-Pilot for Every Journey

Product Blueprint — AI-Powered Travel Management Assistant

Version 1.0

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In One Minute — What Is TripPilot?

Think about the last time you booked a flight. You probably searched two or three websites to compare prices. You booked your ticket. Then a confirmation email landed in your inbox and got buried under a hundred other emails. Later, you had to dig through your inbox to find your PNR number, your seat number, or your terminal.

TripPilot fixes this. It is not a website where you buy flight tickets — airlines and travel sites like MakeMyTrip or Skyscanner already do that. Instead, TripPilot is a smart assistant that sits *on top* of all those websites. It helps you search and compare flights across many sites at once, and once you've booked your ticket anywhere, TripPilot reads your confirmation email or PDF ticket, pulls out all the important details automatically, and keeps everything — every trip, every passenger, every booking — organised in one clean dashboard.

Simple analogy: If airline websites are different shops where you buy travel tickets, TripPilot is your personal travel diary and assistant that remembers everything you bought, from every shop, and answers your questions about it — automatically, without you lifting a finger.

Why the name "TripPilot"?

TRIP Represents the user's entire journey — not just a flight. This lets the product grow later into hotels, trains, buses, visas, insurance, and more.	PILOT Represents the AI assistant that guides the user safely and smartly through their travel, the same way a pilot guides a plane to its destination.
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The name is intentionally future-proof: even though Version 1.0 focuses on flights, "TripPilot" remains an accurate name as the product grows into a full AI travel ecosystem.

Why This App Exists

Modern travellers use multiple platforms to search, compare, and book flights, but once the ticket is booked, all record of it lives scattered across emails, PDFs, screenshots, and different airline apps. There is no single place that brings every trip — regardless of where it was booked — into one organised view. TripPilot exists to close this gap.

What Problem It Solves

TripPilot solves the "travel information chaos" problem. It becomes the single source of truth for a traveller's bookings: it compares prices before booking, and after booking, it automatically imports, reads, and organises every confirmation so the user never has to search their inbox again.

Why People Need It

- Travellers book flights from many different sources and lose track of details.
- Important information like PNR numbers, gate numbers, and baggage rules get buried in emails.
- There is no easy way to see "all my upcoming trips" across different airlines in one screen.
- Comparing prices across multiple sites takes time and manual effort.

How It Is Different From Airline Booking Apps

Airline / OTA Booking Apps	TripPilot
Sell tickets for one airline or one platform only	Does not sell tickets; works across every airline and every booking site
Only show bookings made on their own platform	Imports and organises bookings from any source (email, PDF, boarding pass)
No unified dashboard across providers	One dashboard for all trips, all passengers, all providers
Limited or no AI assistant for trip questions	Built-in AI Travel Assistant that answers natural-language questions

In short: Airline apps help you buy a ticket. TripPilot helps you manage your entire travel life — before, during, and after booking.

02 Problem Statement

Travellers face many small but frustrating problems every time they plan a trip. TripPilot is designed to solve each one directly.

Problem Travellers Face	How TripPilot Solves It
Searching many websites for flights	AI Flight Search checks multiple sources at once and shows combined results
Comparing prices manually	AI Price Comparison ranks flights by price, stops, and duration automatically
Keeping and finding booking emails	Booking Import reads emails/PDFs and saves the details permanently
Losing PNR numbers	PNR is extracted and stored, always visible in the dashboard
Managing multiple trips at once	Reservation Dashboard separates upcoming, current, past, and cancelled trips
Tracking reservation status	Live Flight Tracking shows real-time status such as delayed or boarding
Finding the right terminal or gate	Flight Details store terminal and gate, with alerts if they change
Remembering baggage allowance	Baggage rules are saved and shown per booking
Managing passenger details for every trip	Passenger Management securely stores and reuses traveller information

03 Objectives

1. Allow users to search and compare flights across multiple platforms from one place.
2. Automatically extract booking details from confirmation emails and PDFs using AI and OCR.
3. Provide a single, organised dashboard for all trips, regardless of where they were booked.
4. Securely store and reuse passenger information across multiple bookings.
5. Give users real-time updates on flight status, gate changes, and delays.
6. Offer an AI Travel Assistant that answers travel questions in plain language.
7. Build a system that is secure, reliable, and easy to expand with future travel services.

04 Target Audience

- Business Travellers
- Families
- Students
- Frequent Flyers
- Travel Enthusiasts
- Corporate Employees
- Travel Managers

Audience	Why They Need TripPilot
Business Travellers	Need quick access to flight status, terminal, and gate information while on the move
Families	Need to manage multiple passenger records and bookings together
Students	Book budget flights across many platforms and need everything organised in one place
Frequent Flyers	Track long travel history and reuse saved passenger and preference details
Travel Enthusiasts	Enjoy comparing options and planning trips efficiently
Corporate Employees	Need reliable records of trips for expense and travel reporting
Travel Managers	Manage bookings and passenger details for teams or groups

5.1 AI Flight Search

The user simply types what they want — for example, "Delhi to Mumbai next Friday morning." The AI understands this natural sentence, turns it into a structured search, and queries flight data providers to return matching flights.

5.2 AI Price Comparison

Once results come in from multiple sources, the AI ranks and compares them by price, number of stops, duration, and airline, helping the user pick the best option without manually checking each website.

5.3 Flight Details

Every flight in TripPilot carries a complete, structured set of details so the user never has to look it up elsewhere:

Field	Field
Flight Number	Cabin Class
Airline	Meal
Aircraft Type	Baggage Allowance
Departure / Arrival Airport	Refund Policy
Departure / Arrival Terminal	Cancellation Policy
Departure / Arrival Gate	Seat Availability
Flight Duration	Flight Status
Stops & Layover Details	PNR / Booking Reference
Fare	Booking Website / Airline Name

5.4 Passenger Management

TripPilot securely stores traveller records so they can be reused across future bookings instead of being re-entered every time:

Identity	Contact	Travel Profile
Name, Age, Gender, Passport, Government ID	Phone, Email, Emergency Contact	Frequent Flyer Number, Travel Preferences

5.5 Reservation Dashboard

A single screen organises every trip into clear categories:

- Upcoming Trips
- Current Trips
- Past Trips
- Cancelled Trips
- Travel History
- Statistics

5.6 Booking Import

Instead of booking inside TripPilot, users upload their booking PDF, boarding pass, or confirmation email. The AI reads the document (using OCR and text parsing) and automatically extracts the passenger name, flight number, PNR, seat, dates, and more — then saves it to the dashboard.

5.7 Flight Tracking

TripPilot shows the live status of upcoming flights, including:

- Boarding
- Delayed
- Departed
- Arrived
- Gate Change
- Terminal Change

5.8 Notifications

- Flight reminders before departure
- Gate and terminal update alerts
- Delay and cancellation alerts
- Check-in open reminders

5.9 AI Travel Assistant

A conversational assistant that answers plain-English questions about the user's own trips, for example:

"Find my cheapest flight" · "Show my next journey" · "When should I leave for the airport?" · "Which airline has fewer stops?" · "Show my cancelled trips"

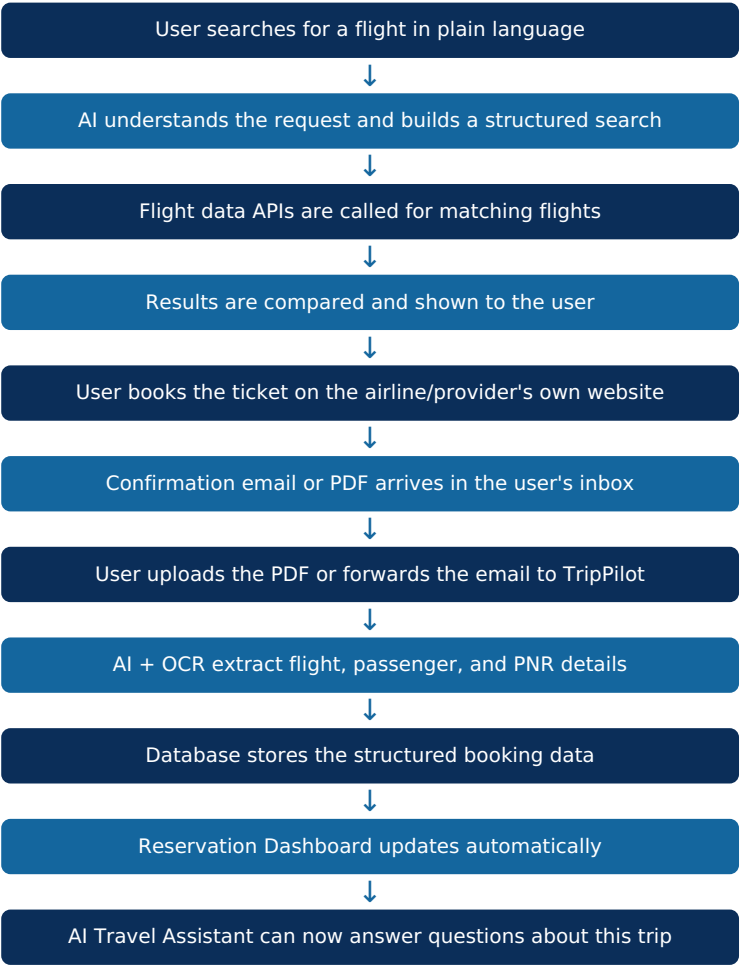
5.10 Admin Dashboard

For the TripPilot team to operate the platform, an internal dashboard provides:

- User Analytics
- Registered Users
- Reservation Statistics
- Popular Routes
- Reports
- System Monitoring

06 Complete User Workflow

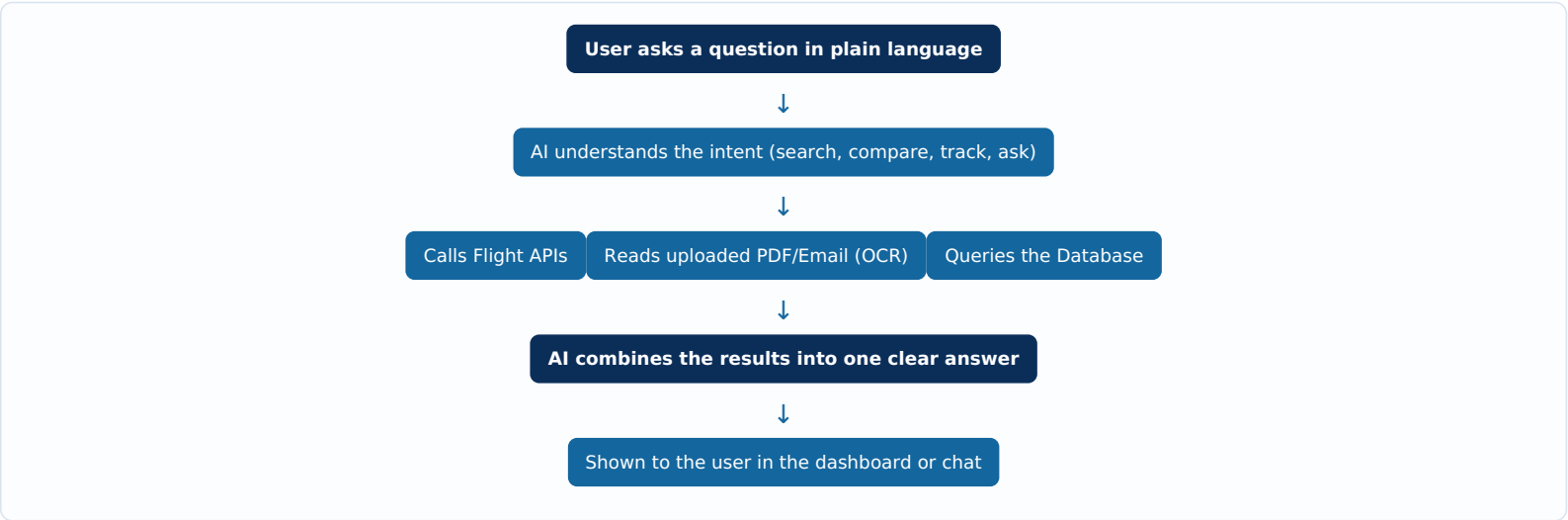
This is the step-by-step journey a user takes inside TripPilot, from searching a flight to getting answers about it later.



Step	What Happens (In Simple Words)
Search	The user types a request like they would text a friend, and the AI turns it into a proper search query.
Compare	TripPilot lays out all matching flights side-by-side so the user can pick the best one.
Book	The actual purchase still happens on the airline or travel website — TripPilot never charges for tickets.
Import	The confirmation the user receives becomes the input TripPilot needs to build a complete trip record.
Organise	Everything is saved neatly so the user can find it anytime, from any device.
Assist	The AI is now aware of every trip and can answer questions instantly.

07 AI Integration

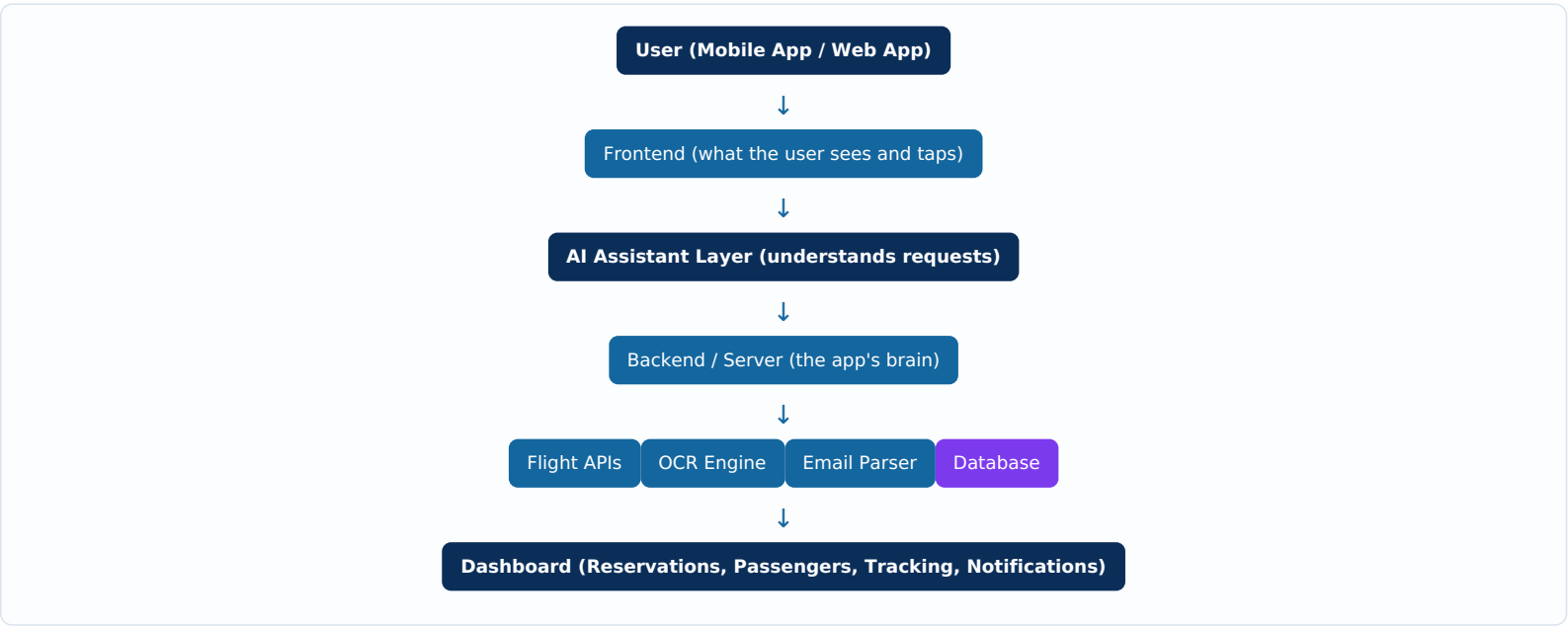
Think of the AI inside TripPilot as a helpful middleman standing between the user and all the technical systems. The user never has to know how flight data is fetched or how a PDF is read — they just talk normally, and the AI handles everything behind the scenes.



Simple explanation: The AI does four things — it *understands* what you want, *fetches* the right information, *stores* it neatly, and *answers* your questions about it. Nothing more complicated than that.

08 System Architecture

System architecture just means "how the different technical parts of the app talk to each other." Here is TripPilot's architecture in simple blocks:



Block	What It Does (Simple Words)
Frontend	The screens and buttons the user interacts with, on phone or web.
AI Assistant Layer	Reads what the user typed and figures out what they actually want.
Backend/Server	The engine that connects everything — frontend, AI, database, and outside services.
Flight APIs	Outside services that provide real flight data (schedules, prices, status).
OCR Engine	Software that reads text from images and scanned PDFs.
Email Parser	A tool that reads the structure of a confirmation email to pull out details.
Database	The storage room where all user, passenger, and booking data lives safely.
Dashboard	The final organised view the user sees, built from everything above.

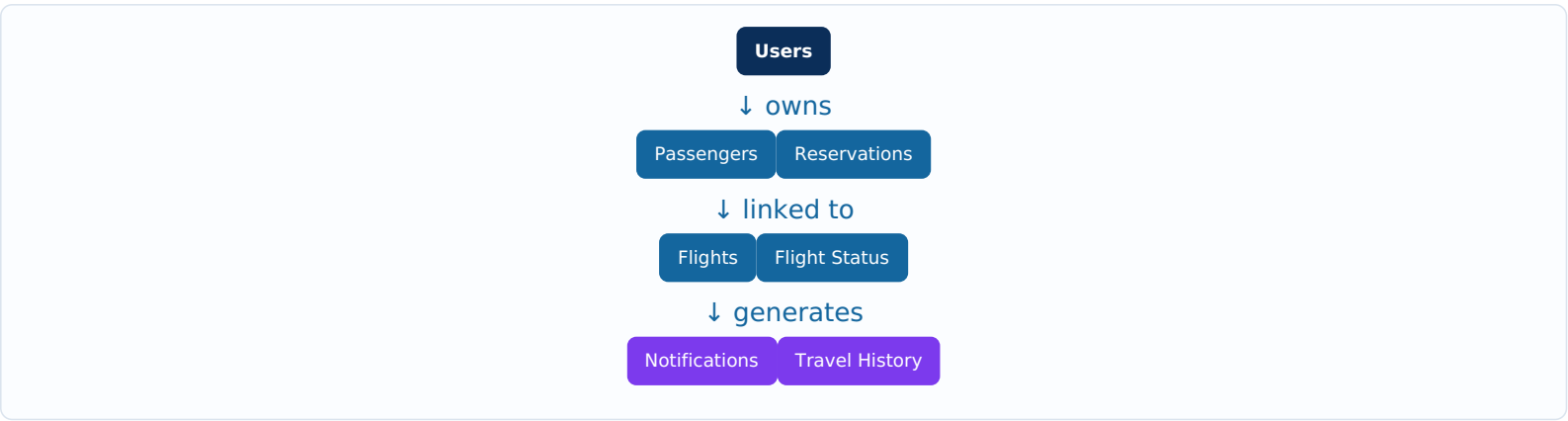
09 Database Design

A database is simply an organised digital filing cabinet. Without it, TripPilot would forget everything the moment the app closes. Each "table" below is like one drawer in the cabinet, dedicated to one type of information.

Suggested Tables

Table	Stores
Users	Login details, account info
Passengers	Traveller identity and contact details
Reservations	Booking records linking users, passengers, and flights
Flights	Flight number, airline, route, timing, fare
Flight Status	Live status updates: delayed, boarding, etc.
Notifications	Reminders and alerts sent to users
Travel History	Archive of completed and past trips

Simple Relationship Diagram



In simple words: one User can have many Passengers and many Reservations. Each Reservation is linked to one Flight, and each Flight has a live Flight Status. Reservations that are complete move into Travel History, and important updates create Notifications.

10 APIs

An API (Application Programming Interface) is simply a way for two different computer systems to talk to each other. Imagine ordering food through a waiter: you tell the waiter what you want, the waiter goes to the kitchen, and comes back with your food. An API is that waiter — TripPilot asks a flight data provider for information, and the API brings the answer back.

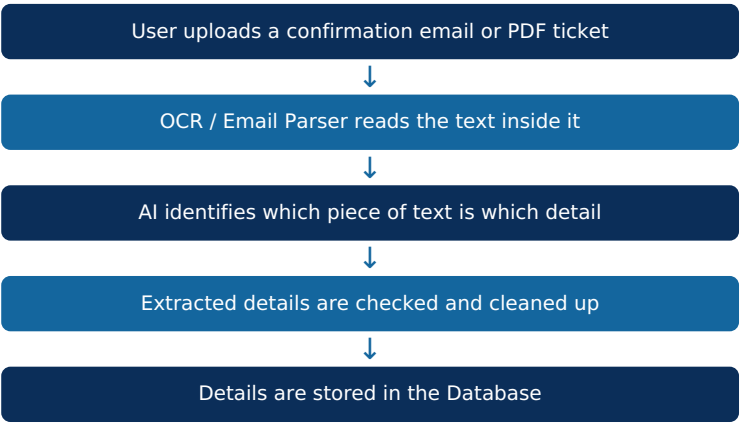
Why Use APIs Instead of Collecting Flight Data Manually?

- Flight schedules, prices, and seat availability change every minute — a human cannot track this manually.
- APIs give accurate, real-time data directly from trusted sources.
- It saves enormous development time compared to building this data from scratch.

Examples of flight data providers (mentioned only as examples): Amadeus, Skyscanner. TripPilot can integrate with one or more such providers depending on cost and coverage.

11 OCR & Email Parsing

OCR stands for "Optical Character Recognition." In simple words, it is technology that lets a computer *read* text from a picture or scanned document, the same way a human eye reads a printed page.



Details Extracted

- PNR
- Passenger Name
- Flight Number
- Airline
- Seat
- Terminal
- Date
- Time

12 Algorithms Used In TripPilot

An algorithm is simply a set of step-by-step instructions to solve a problem — like a recipe for a computer. Below are the key algorithms TripPilot relies on, explained in plain words.

Binary Search

What it is: A fast way to find something in a sorted list by repeatedly checking the middle item and eliminating half the list each time.

Why used: It is much faster than checking every item one by one.

Where used in TripPilot: Quickly finding a specific flight or booking in a large, sorted list of results.

Merge Sort

What it is: A sorting method that splits a list into smaller pieces, sorts each piece, then merges them back together in order.

Why used: It stays reliably fast even with very large lists.

Where used in TripPilot: Sorting large sets of flight search results by price or duration.

Quick Sort

What it is: A sorting method that picks a "pivot" value and arranges smaller items on one side and larger items on the other, then repeats this on each side.

Why used: It is usually very fast in practice for everyday-sized lists.

Where used in TripPilot: Sorting flights shown in the price comparison screen.

Hash Maps

What it is: A storage structure that lets the computer jump straight to a piece of data using a unique key, instead of searching through everything.

Why used: It gives near-instant lookups.

Where used in TripPilot: Instantly looking up a booking using its PNR number.

Queue

What it is: A line where the first item added is the first one processed — just like a queue at an airport counter.

Why used: It keeps tasks processed fairly, in the order they arrived.

Where used in TripPilot: Processing uploaded booking PDFs one after another in the order they were received.

Priority Queue

What it is: Similar to a queue, but the most important item is processed first, regardless of when it arrived.

Why used: Some tasks are more urgent than others.

Where used in TripPilot: Sending an urgent "gate change" alert before a routine reminder notification.

Graphs

What it is: A way of representing connections between things — like cities connected by flight routes.

Why used: Flight networks are naturally made of connected points (airports) and paths (routes).

Where used in TripPilot: Representing airports and the flight routes that connect them, especially for journeys with layovers.

Dijkstra's Algorithm

What it is: A method for finding the shortest (or cheapest, or fastest) path between two points in a graph.

Why used: It finds the most efficient route without checking every possible option manually.

Where used in TripPilot: Suggesting the fastest or cheapest multi-stop route between two cities.

Big O Notation

What it is: A simple way to describe how much slower a program gets as the amount of data grows — it measures efficiency, not exact speed.

Why used: It helps developers pick the right algorithm before the app has millions of users.

Where used in TripPilot: Used during development to make sure search, sorting, and lookup features stay fast even as the number of flights and users grows.

13 Recommended Technology Stack

Layer	Recommended Technology
Frontend	React / React Native (web and mobile)
Backend	Node.js with Express, or Python with FastAPI
Database	PostgreSQL (structured data) with Redis for fast caching
Authentication	OAuth 2.0 / JWT-based secure login
AI	Large Language Model API (e.g. Claude) for understanding and answering queries
OCR	Tesseract OCR or a cloud OCR service
Cloud	AWS, Google Cloud, or Microsoft Azure
Notifications	Firebase Cloud Messaging / email & SMS gateway
Analytics	Google Analytics / Mixpanel
Hosting	Cloud-based container hosting (e.g. AWS ECS, Google Cloud Run)
Deployment	Docker containers
CI/CD	GitHub Actions or GitLab CI
Monitoring	Grafana, Prometheus, or a cloud-native monitoring tool

14 Security

Area	Approach
Authentication	Secure login with hashed passwords and optional two-factor authentication
Encryption	Data encrypted both in transit (HTTPS) and at rest in the database
Secure APIs	All API calls authenticated and rate-limited to prevent abuse
Privacy	Passenger data (passport, ID) treated as highly sensitive and access-restricted
Data Protection	Compliance with relevant data protection regulations
Database Backups	Regular automated backups to prevent data loss
Role-Based Access	Different access levels for regular users, travel managers, and admins

15 Future Features

Because "Trip" was chosen deliberately over "Flight," TripPilot is designed to grow well beyond air travel:

- Hotel Management
- Train Tickets
- Bus Tickets
- Visa Tracking
- Travel Insurance
- Currency Converter
- Expense Manager
- Calendar Sync
- Offline Tickets
- AI Itinerary Planner
- Travel Recommendations

16 Development Roadmap



17 AI Assisted Development

Modern AI tools can significantly speed up how TripPilot is built. Tools like Claude, ChatGPT, GitHub Copilot, Cursor, Windsurf, Replit AI, and similar assistants can help generate code, write documentation, and suggest fixes far faster than working entirely by hand.

Important: These tools accelerate development, but they do not replace it. Human developers must still review the code, test it thoroughly, and manage deployment to make sure the product is safe, correct, and reliable before real users depend on it.

Closing Note

This blueprint captures the vision, features, architecture, and roadmap for **TripPilot Version 1.0** — an AI-powered travel management assistant built to bring order to the traveller's scattered bookings. It is intended as a working reference for developers, designers, and stakeholders before development begins.